

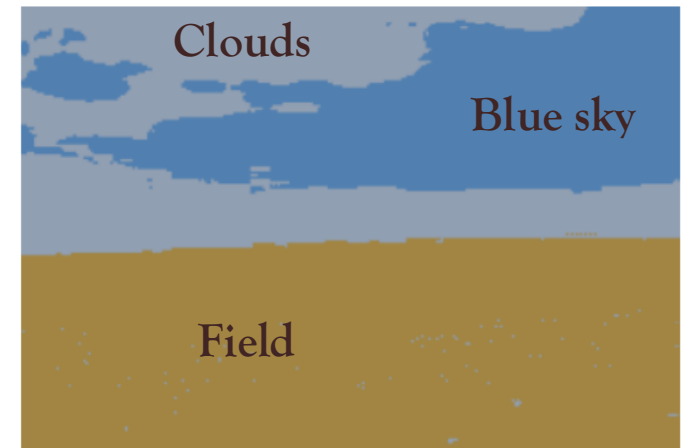


4. IMAGE SEGMENTATION

1. *Histogram-based segmentation*
2. *Clustering-based segmentation*
3. *Region growing and merging*
4. *Tagging*
5. *Morphological transformations*

TAGGING

- From segmentation (or from image) → TAGS
- Textual **descriptions** of each region



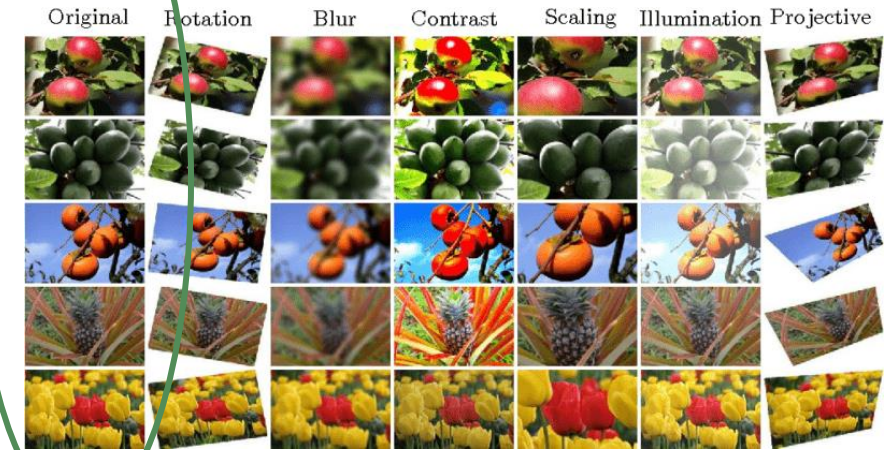
TAGGING: METHODS

- **Manually:** go through images and/or segmentations and assign tags. Challenges:
 - **Inconsistent** if multiple annotators disagree → label images several times + decision rule (e.g., majority voting)
 - **Time-consuming** (including annotators training process)
 - **Expensive** and difficult to scale to large datasets
- **Semi-automated methods:** provide a starting point (e.g., suggested initial tags) for the annotator. ↓time-consuming

TAGGING: METHODS

- Fully automated using **synthetic data** (synthetic images):
 - Automatically **generate** images that mimic real data
 - The **labels/tags** are **known in advance**: the operator controls what it is generated
 - Accurate and cost-effective
 - Methods:
 - Data augmentation
 - **Variational Autoencoders (VAE)**: start from existing data, create a new data distribution, and map it back to the original space using an encoder-decoder method
 - **Generative Adversarial Networks (GAN)**: model with 2 neural networks that compete against each other. One neural network attempts to create **fake images**, while the other tries to distinguish real / fake. At the end, the system becomes able to generate fake images hard to distinguish from real ones
 - Etc.

Tagged



<https://research.aimultiple.com/data-augmentation/>

Tags: faces / people



<https://thispersondoesnotexist.com/>
StyleGAN (Karras et al.)

MORPHOLOGICAL OPERATIONS

- For binary images (segmentations/**masks**)
- Post-segmentation **image processing** to improve **quality** of segmentation
- Fill holes, eliminate isolated pixels (noise filtering), smooth boundaries, etc.

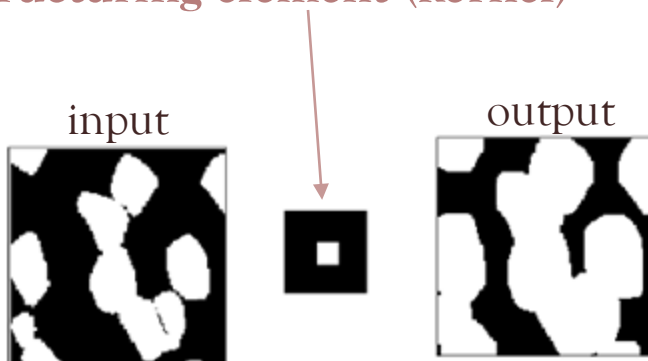
Note that in the following examples sometimes the foreground is *white* and sometimes it is *black*

Image after segmentation



MORPHOLOGICAL OPERATIONS

- Interaction of masks with the **structuring element (kernel)**



- Erosion / dilation
- Opening /closing

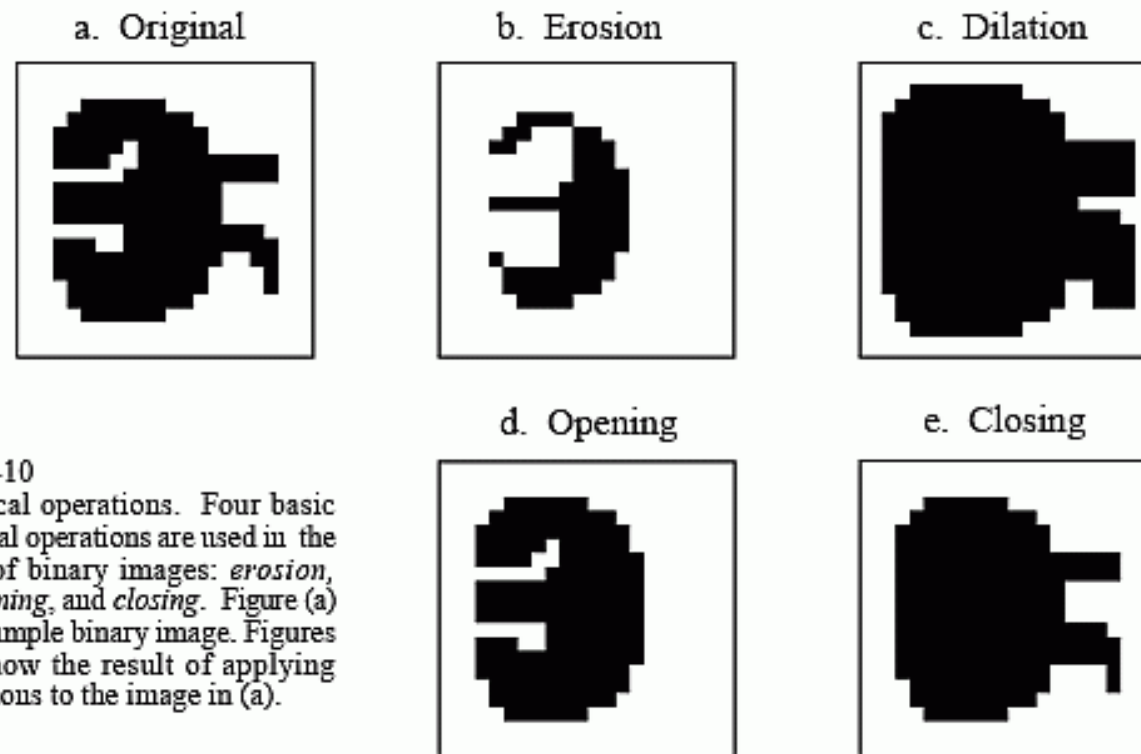
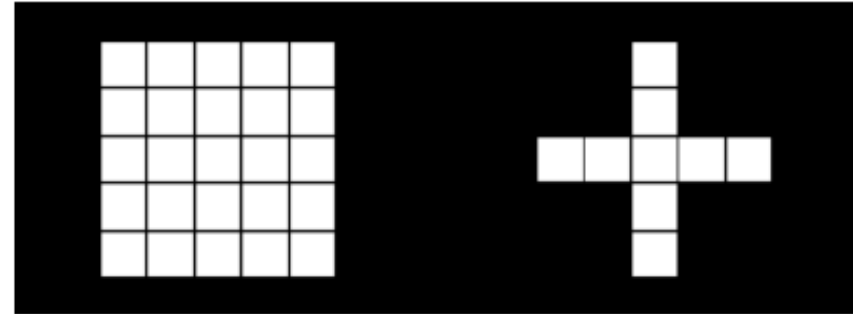


FIGURE 25-10
Morphological operations. Four basic morphological operations are used in the processing of binary images: *erosion*, *dilation*, *opening*, and *closing*. Figure (a) shows an example binary image. Figures (b) to (e) show the result of applying these operations to the image in (a).

The Scientist and Engineer's Guide to Digital Signal Processing By Steven W. Smith

MORPHOLOGICAL OPERATIONS

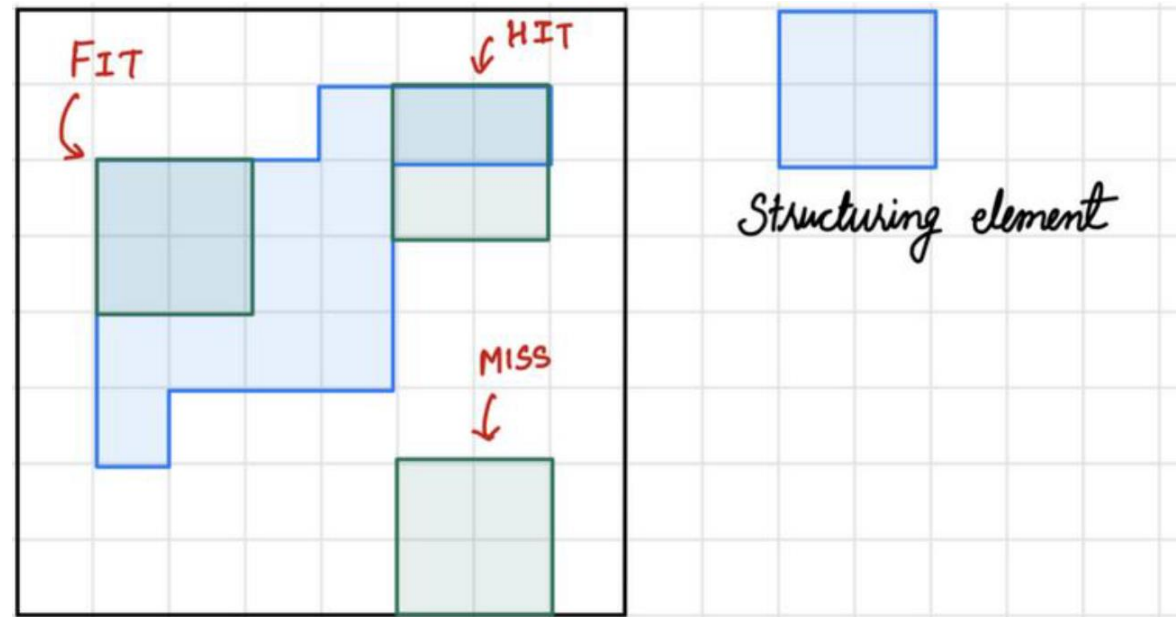
- The most important factor is the **structuring element (kernel)**
- Small matrix composed of binary pixels
- Any **shape**
- Most common: square, cross (various sizes)



- Its shape determines the output (post-processed mask)
- It is positioned at all possible locations in the image & compared with the mask
- Similar to **spatial filtering**

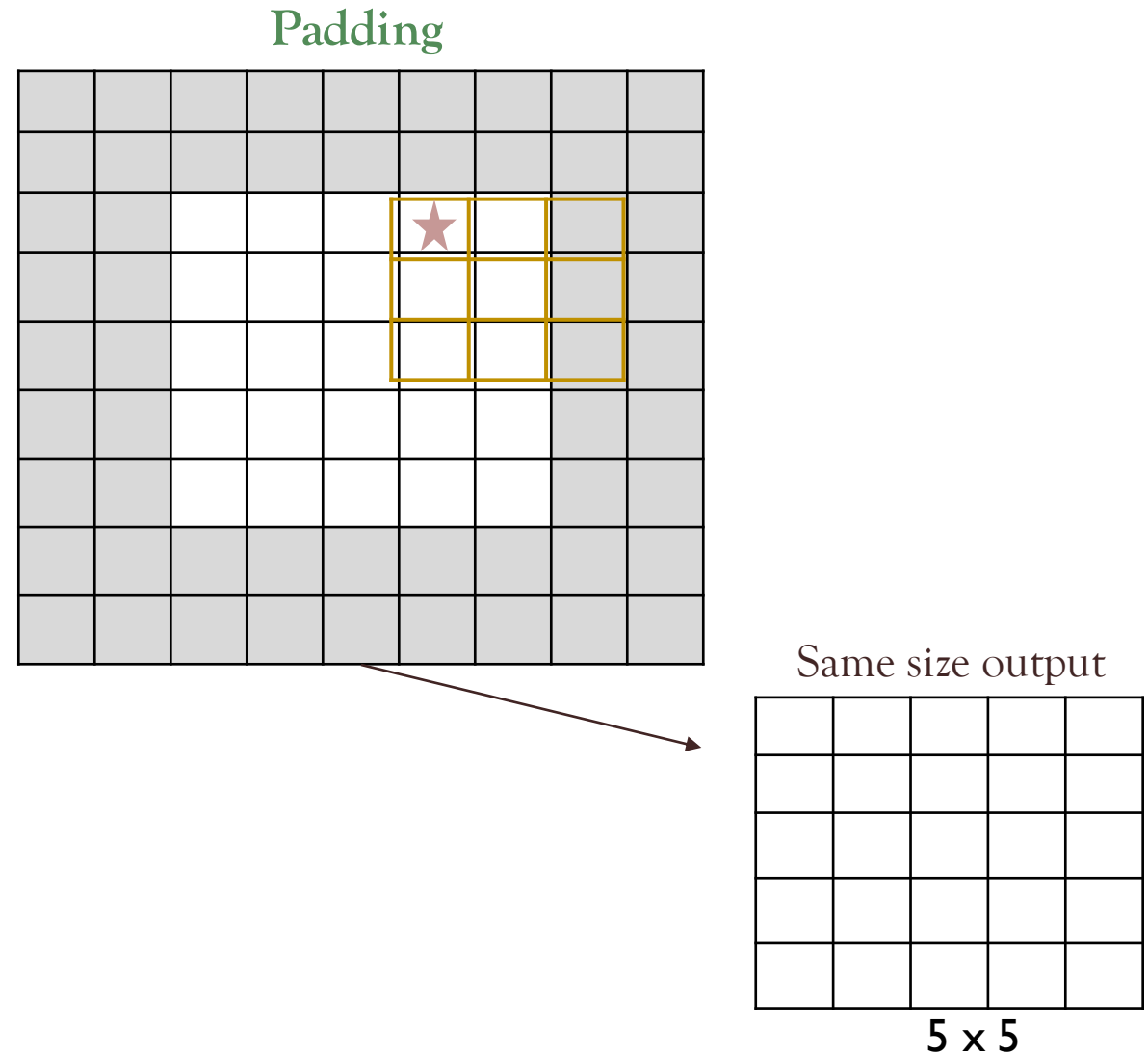
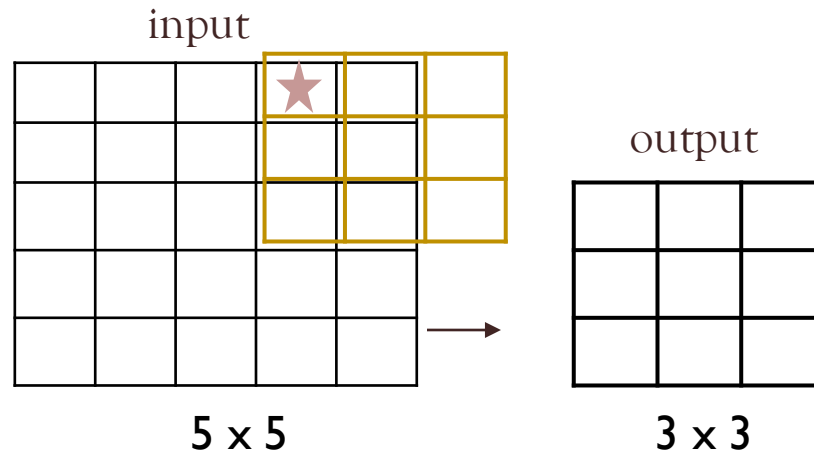
MORPHOLOGICAL OPERATIONS

- **Fit:** all pixels in the structuring element overlap with the pixels of the object
- **Hit:** at least one pixel in the structuring element overlaps with the object
- **Miss:** No pixel in the structuring element overlaps the pixels of the object



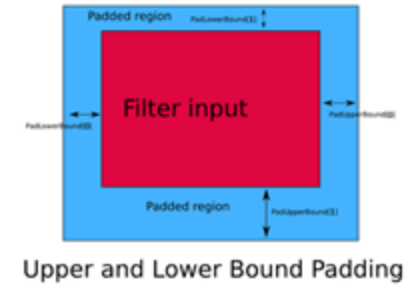
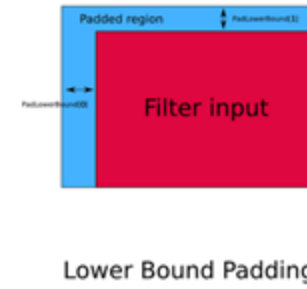
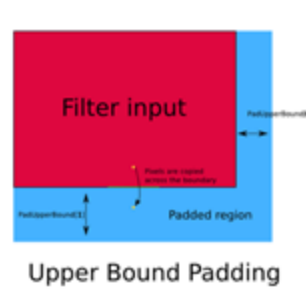
Chhikara P, 2022

MORPHOLOGICAL OPERATIONS: BORDERS?



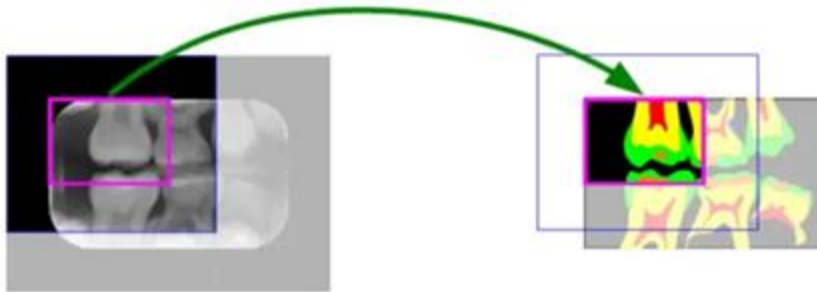
MORPHOLOGICAL OPERATIONS: PADDING

- Symmetric / non-symmetric



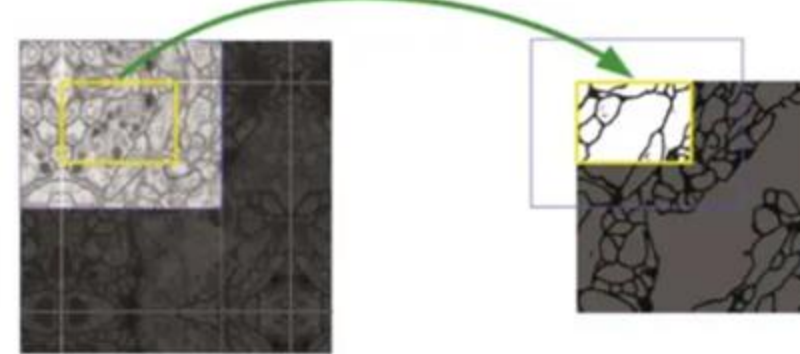
https://itk.org/Doxygen/html/classitk_1_1MirrorPadImageFilter.html

Zero-padding



Review: U-Net (Biomedical Image Segmentation). Towards Data Science

Mirror-padding (continuity preservation)

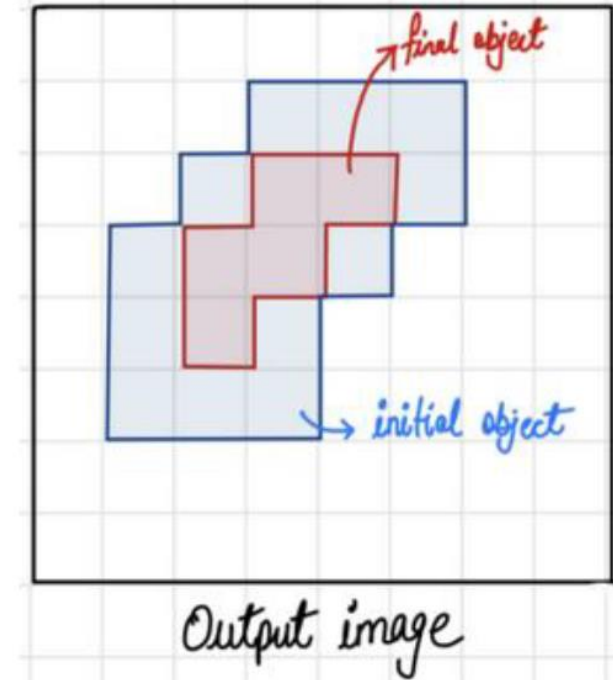
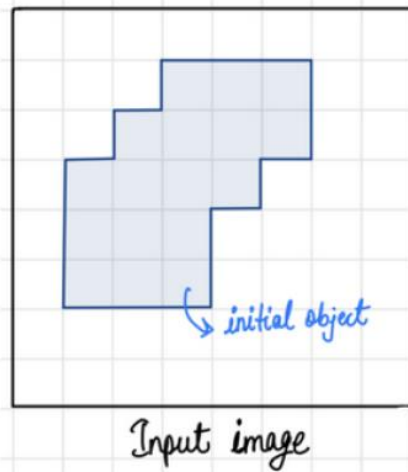
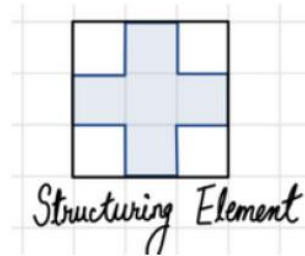


- Number of added rows and columns depends on the kernel's size

MORPHOLOGICAL OPERATIONS: EROSION

output $\begin{cases} 1 \text{ if FIT} \\ 0 \text{ otherwise} \end{cases}$

FIT: **all** pixels overlap

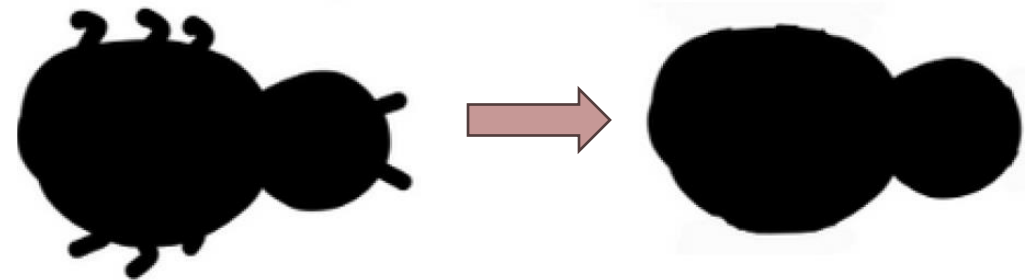
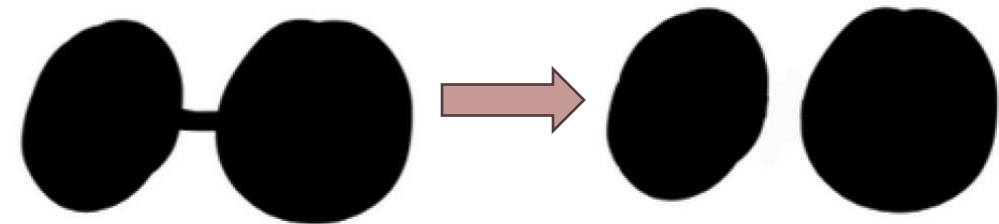
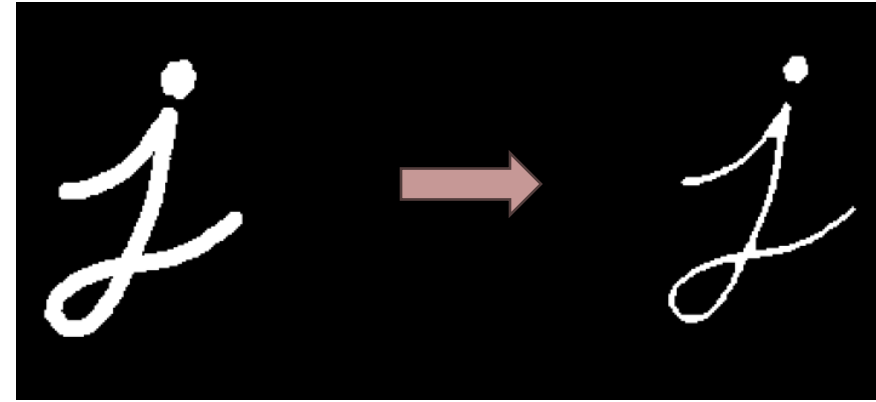


Chhikara P, 2022

MORPHOLOGICAL OPERATIONS: EROSION

output $\begin{cases} 1 & \text{if FIT} \\ 0 & \text{otherwise} \end{cases}$

- It removes image pixels on the boundaries, **shrinking** the object
- It can **split apart** joint objects
- It can strip away **extrusions**

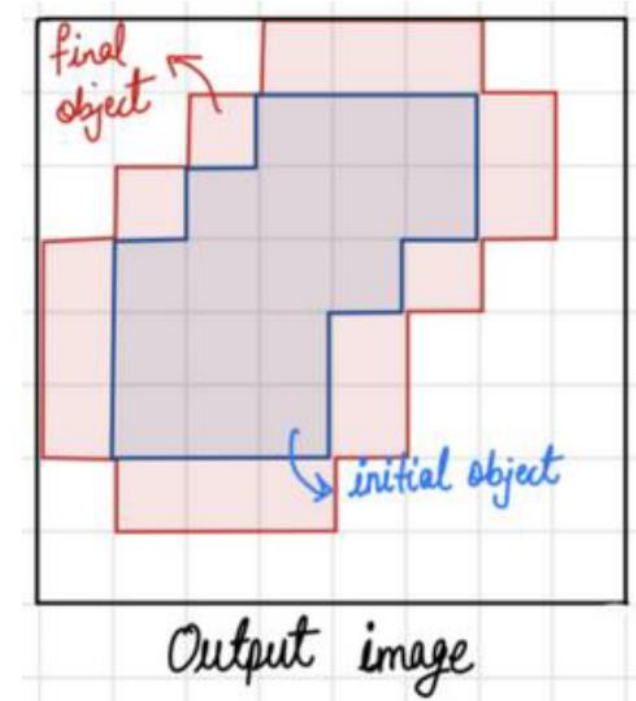
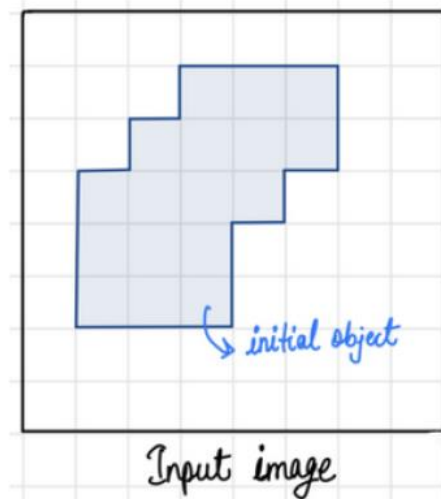
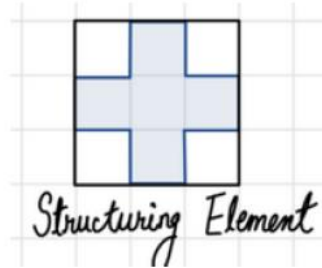


Chhikara P, 2022

MORPHOLOGICAL OPERATIONS: DILATION

output $\begin{cases} 1 \text{ if HIT} \\ 0 \text{ otherwise} \end{cases}$

HIT: at least 1 pixel overlaps

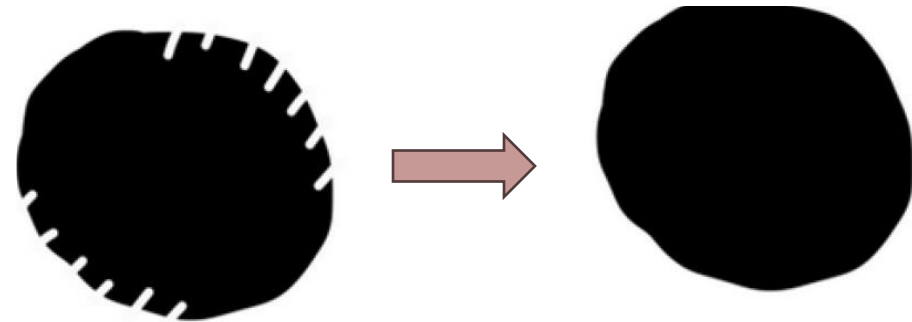
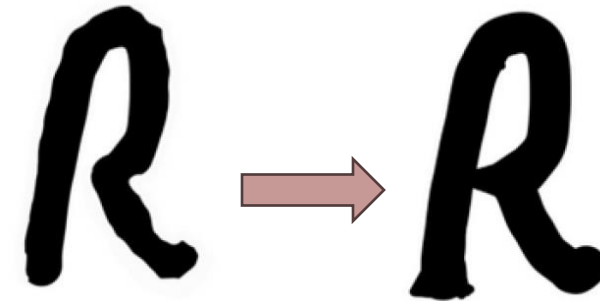
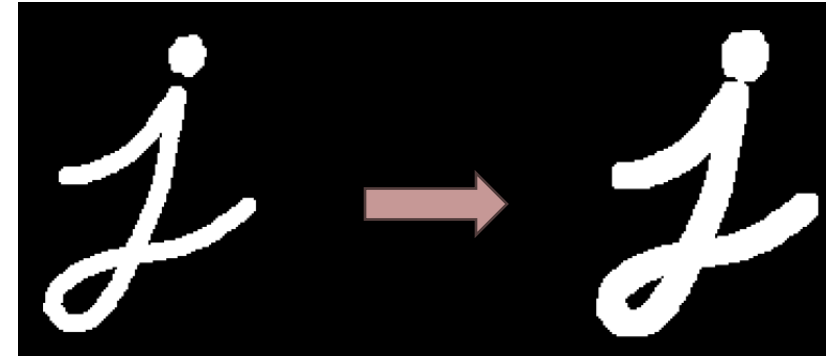


Chhikara P, 2022

MORPHOLOGICAL OPERATIONS: DILATION

output $\begin{cases} 1 & \text{if HIT} \\ 0 & \text{otherwise} \end{cases}$

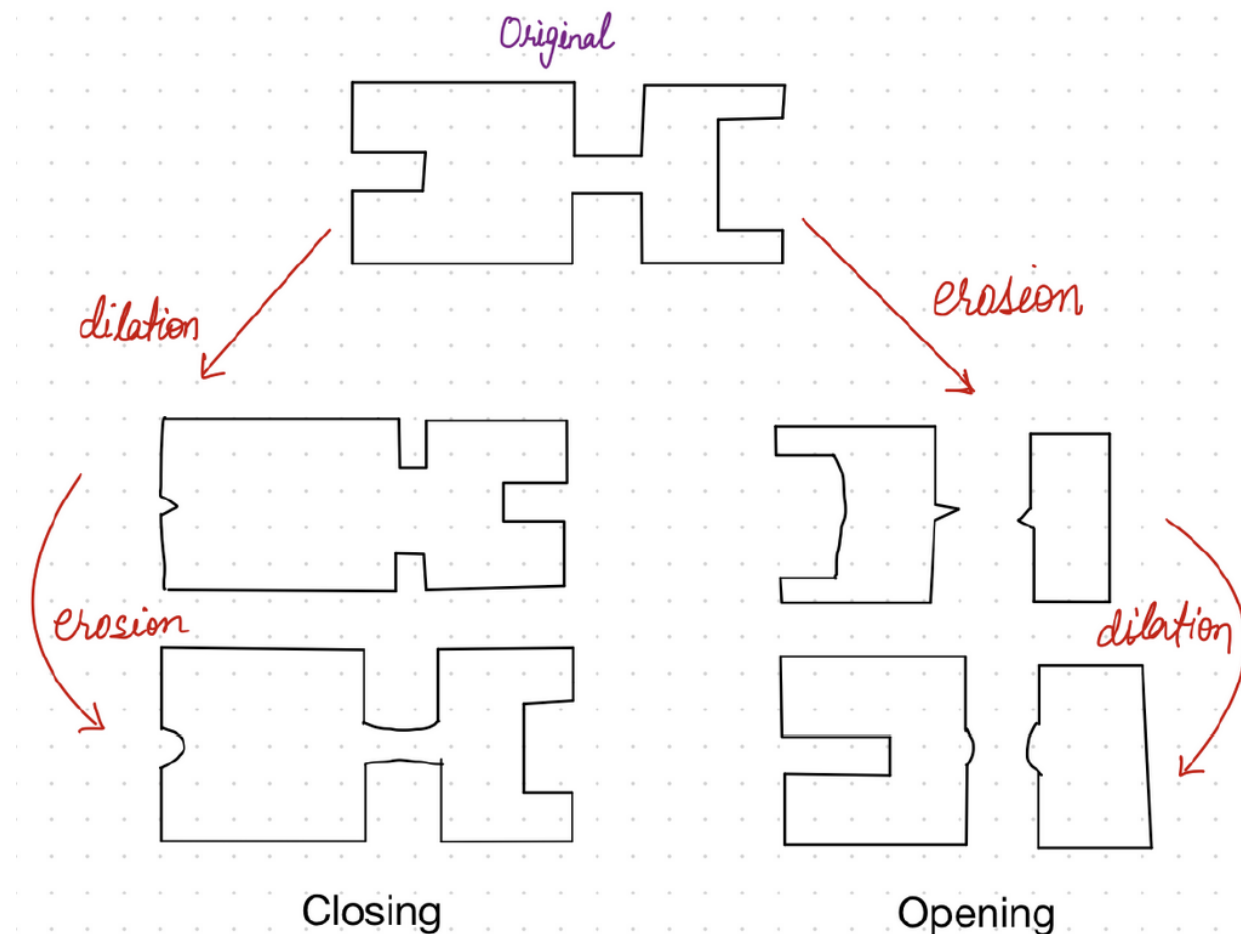
- It **expands** the object, adding pixels on the object's boundaries
- It can correct **breaks**
- It can correct **intrusions**



Chhikara P, 2022

MORPHOLOGICAL OPERATIONS

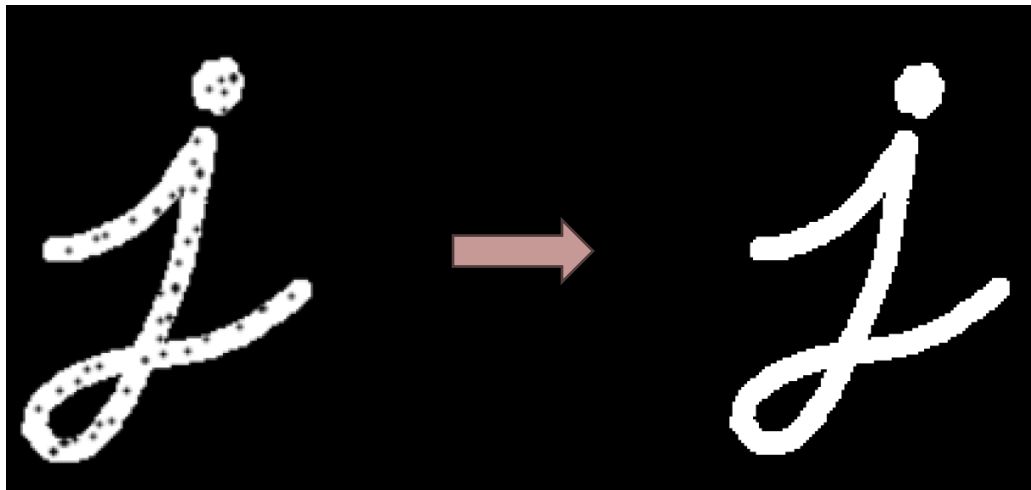
- **Compound operations:** combination of other morphological operations
- Most widely used:
 - **Closing** → first dilation, then erosion
 - **Opening** → first erosion, then dilation



Chhikara P, 2022

MORPHOLOGICAL OPERATIONS

Closing



- Gap filling, like dilation, but **without** modifying the boundary
- Noise filtering

Opening



- Erase spurious pixels, like erosion, but **without** modifying the boundary
- Noise filtering

MORPHOLOGICAL OPERATIONS

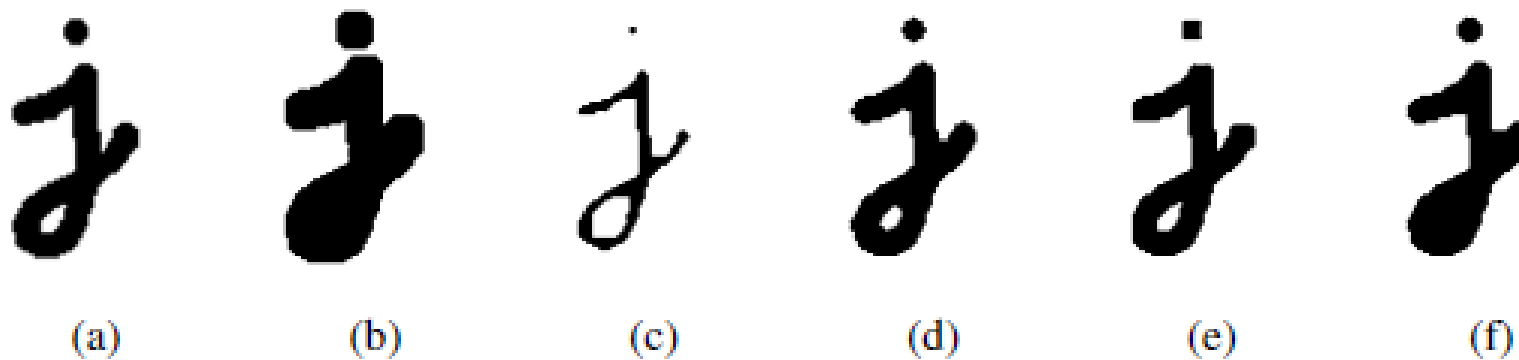


Figure 3.21 Binary image morphology: (a) original image; (b) dilation; (c) erosion; (d) majority; (e) opening; (f) closing. The structuring element for all examples is a 5×5 square. The effects of majority are a subtle rounding of sharp corners. Opening fails to eliminate the dot, since it is not wide enough.

Szeliski, R. (2022). Computer vision: algorithms and applications. Springer Nature.

* majority: operation in between fit & hit that results in 1 if more than half (the majority) of the voxels overlap, and 0 otherwise

MORPHOLOGICAL OPERATIONS

- **Edge extraction** → first erosion, then subtract initial mask



Chhikara P, 2022

MORPHOLOGICAL OPERATIONS

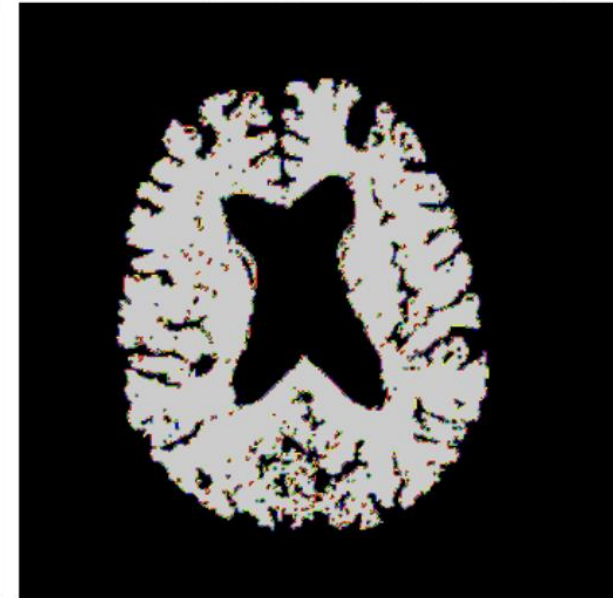
- Python code → Lab tomorrow



GM segmentation



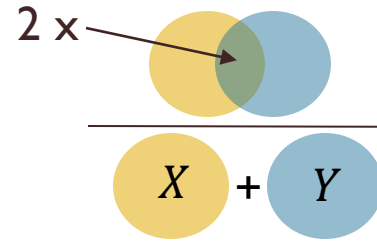
After morphological postprocessing



SEGMENTATION PERFORMANCE EVALUATION

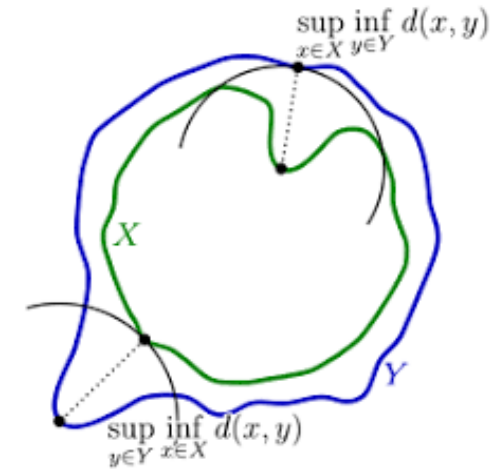
1. Based on overlap precision: Sørensen–Dice similarity coefficient

$$Dice = \frac{2|X \cap Y|}{|X| + |Y|} = \frac{2 TP}{2 TP + FP + FN}$$



X : estimation
 Y : GT

- If perfect overlap: Dice = 1
 - If no overlap at all: Dice = 0
2. Based on the **precision of the contours**:
 - Average symmetric surface distance (ASSD) = $\text{mean}\{\text{dist}(X, Y), \text{dist}(Y, X)\}$
 - Hausdorff distance (HD) = $\max\{\max_dist(X, Y), \max_dist(Y, X)\}$
 - Good results if $\downarrow$ ASSD, $\downarrow$ HD



Wikipedia

SEGMENTATION PERFORMANCE EVALUATION

3. **Classification metrics** → image segmentation can be seen as classification at the pixel/voxel level
- confusion matrix, accuracy, precision, recall, F1-score, etc.

		Predicted label	
		Positive	Negative
True label	Positive	TP	FN
	Negative	FP	TN

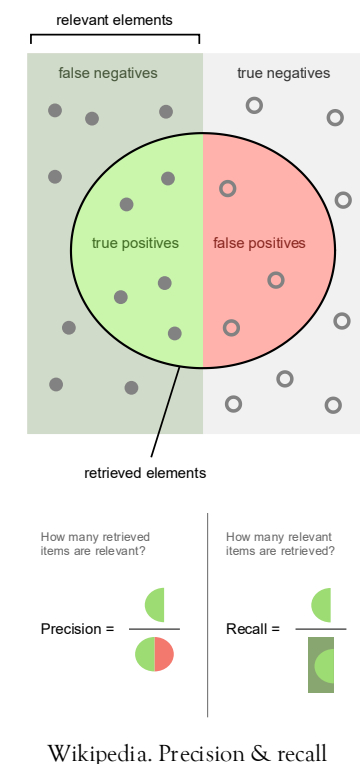
TP: True Positive

TN: True Negative

FP: False Positive

FN: False Negative

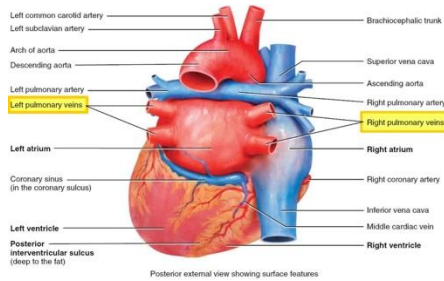
- $Accuracy = \frac{TP+TN}{all}$
- $Sensitivity(\textit{Recall or TP rate}) = \frac{TP}{P} = \frac{TP}{TP+FN}$
- $Precision(\textit{Specificity or TN rate}) = \frac{TN}{N} = \frac{TN}{TN+FP}$
- $F1_score = 2 * \frac{Precision \cdot Recall}{Precision+Recall} = \frac{2TP}{2TP+FP+FN}$
- Precision-recall curve & area under the curve



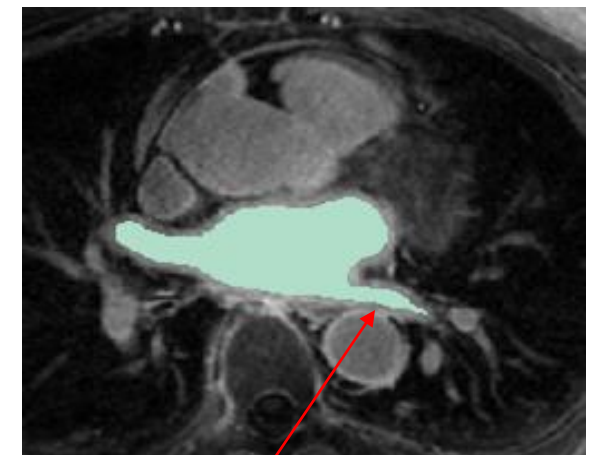
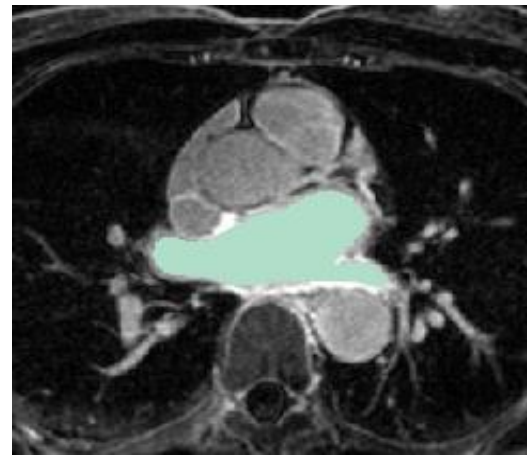
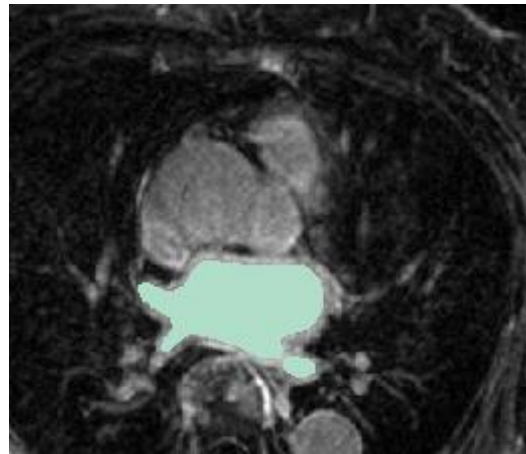
PERFORMANCE EVALUATION: BEST METRIC?

- Problem-specific compromise...
- **Domain expertise** is crucial

Left atrium + Pulmonary Veins (PV) segmentation



<https://healthjade.net/pulmonary-vein/>



If inaccurate PV segmentation:

- $\uparrow$ Dice (overlap) (good)
- $\uparrow$ ASSD, $\uparrow$ HD (distance) (bad)

Is accurate PV segmentation needed?

Maier-Hein, L., & Menze, B. (2022). **Metrics reloaded: Pitfalls and recommendations for image analysis validation.** Nature methods, 2024

EXAMPLE: BRAIN VENTRICLES FROM MRI

- Semi-automatic segmentation with ITK-SNAP
- Steps:
 1. Define Region Of Interest (ROI)
 2. Preliminary intensity threshold (or edges/contours, etc.)
 3. Seed-based region growing (avoiding edges!)

